

Ex no: 1

DEVELOP AN APPLICATION THAT USES GUI COMPONENTS,
FONT AND COLORS

Aim:-

TO develop a simple application using GUI components that allows the user to change font style, size & color of text

ALGORITHM:-

- 1) Start the Program
- 2) Import required GUI libraries
- 3) Create the main window
- 4) Add a text component (label/textbox)
- 5) Add options for:-
 - Font Style
 - Font size
 - Font color
- 6) Add buttons to select font and color.
- 7) When user select options:
 - update the text appearance dynamically
- 8) Display the modified text on screen.
- 9) End the Program.
- 10) Run the android application.

RESULT:

Thus a simple android application that uses GUI components, Font and colors is developed successfully.

EX NO: 2

DEVELOP AN APPLICATION THAT USES LAYOUT MANAGERS AND EVENT LISTENERS

AIM:-

TO develop a simple android application that uses layout managers and event listeners.

ALGORITHM:

- 1) open android studio and start a new project called "ex no:2" with empty activity.
- 2) open activity_main.xml in res > layout.
- 3) Add textviews, edittexts, spinner and button in design tab.
- 4) Edit the xml in text tab if needed.
- 5) code button actions in MainActivity.java.
- 6) Run the app

RESULT:

thus a simple android application that uses layout managers and event listeners is developed successfully.

Aim:-

To develop a mobile application in application creating android studio that allows the user to send an email using an email intent.

ALGORITHM:-

- 1) Start the program.
- 2) Open android studio and create a new project.
- 3) Choose an empty activity and configure the project.
- 4) Design the user interface with input fields:
 - Recipient email
 - Subject
 - message body
- 5) Add a "send email" button in the layout.
- 6) Initialize UI components in the MainActivity file.
- 7) Set an OnClickListener for the send button.
- 8) Create an intent with action ACTION_SEND
- 9) Pass email details using intent extras.
- 10) Launch the email application using startActivity() and end the program.

RESULT!

Thus a simple android application that uses to allow the user to send an email is developed successfully.

Ex no: 4

WRITE AN APPLICATION THAT DRAWS BASIC
GRAPHICAL PRIMITIVES ON THE SCREEN

Aim:-

to develop an android application to draw basic graphical primitives.

ALGORITHM:-

1. open android studio
2. create a new empty activity project
3. design the layout and add a custom view to xml file
4. use canvas and paint methods in onDraw() to draw line, rectangle & circle.
5. Run the application on an emulator after create a new Java class that extends view.
6. Display the graphical primitives.

RESULT:

Thus a simple android application that draws basic graphical primitives on the screen is developed successfully.

Ex no: 6

DEVELOP A APPLICATION THAT MAKE USE OF
DATABASE

AIM:-

to develop a simple android application that makes use of database

ALGORITHM:

- > Open Android studio and create a new project using Empty Activity.
- * Add Internet permission in AndroidManifest.xml.
- * Design the user interface using TextView & ScrollView.
- * Write code to fetch and parse the RSS feed using Xml Parser.
- * Display the RSS feed content on the screen.

RESULT:

Thus an android application that makes use of RSS feed is developed successfully.

Ex no: 6

DEVELOP A MOBILE APPLICATION "VOICE-TO-TEXT
WITH EXTERNAL AUDIO INPUT DEVICES"

AIM:-

to develop a mobile application in android studio that converts speech from an external audio i/p device into text.

ALGORITHM:-

1. Start the program.
2. Open android studio and create a new project.
3. Add audio permission and connect to the external audio input device (microphone).
4. Design the interface with a TextView and a Button.
5. On button click, capture voice input using speech recognition and convert it to text.
6. Display the converted text on the screen and stop the program.

RESULT:

Thus a simple android application that convert the speech into text successfully.

Ex no: 7

IMPLEMENT AN APPLICATION THAT CREATES
AN ALERT RECEIVING A MESSAGE

Aim:-

To develop an android application that creates an alert upon receiving a message

- 1) Start the program
- 2) open android studio and create a new project
- 3) start the application, Register a broadcast receiver for incoming message.
- 4) Detect incoming SMS
- 5) Display the alert message to the user.
- 6) Stop the application

RESULT:

Thus the application was successfully developed to receive SMS messages and display an alert notification on screen whenever a new message received.

Ex no: 8

DEVELOP A MOBILE APP THAT CONNECTS TO A WEARABLE FITNESS TRACKER OR A HEART RATE MONITOR VIA BLUETOOTH

Aim:-

TO develop an android application that connects to a wearable fitness tracker or heart rate monitor using bluetooth and display the heart rate data.

Algorithm:-

- 1) create a new android project in android studio.
- 2) Add bluetooth permissions in androidmanifest.xml.
- 3) Design UI in activity_main.xml (buttons & TextView)
- 4) initialize bluetoothAdapter in MainActivity.java.
- 5) Enable Bluetooth and connect to the selected device
- 6) Receive heart rate data and display it in TextView

RESULT:

The application was successfully developed to connect with a wearable fitness tracker via bluetooth and display the heart rate data in real-time

Ex no: 09

DESIGN A MOBILE APP TO "CAPTURING AND PROCESSING AUDIO WITH EXTERNAL MICROPHONES"

Aim:-

TO develop an android application that captures audio using an external microphone and processing the recorded audio data

ALGORITHM:

- 1) Create a new project in Android Studio
- 2) Add RECORD - AUDIO permission in AndroidManifest.xml.
- 3) Design UI in activity-main.xml
- 4) Initialize MediaRecorder/AudioRecord in MainActivity.java
- 5) Start and Stop audio recording using external microphone
- 6) Process recorded audio and display the result.

RESULT:

The application was successfully developed to capture audio using an external microphone and process the record audio & data efficiency.

Eno: 10

Electric VEHICLE (EV) CHARGING STATION LOCATOR

Aim :-

To develop an android application that locates nearby electric vehicle charging station using location

ALGORITHM:

- 1) Create a new project in Android studio.
- 2) Add Permissions for location and internet in AndroidManifest.xml
- 3) Integrate Google maps API in the application
- 4) Design UI to display map and user location.
- 5) Fetch nearby EV charging station data using location Services / API
- 6) Display charging stations as markers on the map

RESULT:

The application was successfully developed to locate and display nearby EV charging stations on a map.